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PRACTICE 05. – WEB PROGRAMMING

1. **Pixel-art** Create a pixel-art editor. When we draw a pixel-art image, we color the cells of a grid for different colours. The page will have several parts. The first will be the grid itself, where we can click on a cell to be coloured. Right-clicking a cell we can delete the colouring. The second part will be a colour picking part, where a new color can be selected, and it contains a list of the previously chosen colours. The third part will make it possible to manage several pixel-art images. This is a list which contains the previews of the pixel-art images, and by choosing one of them, that pixel-art is loaded in the editor grid. After editing we can save them back to the list. The list elements are automatically persisted in the local storage layer of the browser.

```
<table class="preview">
  <tr>
    <td style="background-color: #ff0000"></td>
    <td style="background-color: #00ff00"></td>
    <td style="background-color: #0000ff"></td>
  </tr>
</table>
<hr>
<table class="edit">
  <tr>
    <td style="background-color: #ff0000"></td>
    <td style="background-color: #00ff00"></td>
    <td style="background-color: #0000ff"></td>
  </tr>
</table>
```

```

table.preview, table.edit {
  border-collapse: collapse;
}
table.preview td {
  border: none;
  width: 3px;
  height: 3px;
  padding: 0;
}
table.edit td {
  border: 1px solid lightgray;
  width: 20px;
  height: 20px;
  padding: 0;
}

```

- I. Generating the grid
 - a. Create a form on which the width and height of the grid (table) can be given!
 - b. Click a button to display the grid with the selected dimensions! If you want, use the `edit` style class specified above!
 - c. *Set a minimum and a maximum value for the input fields. If a bad value is written, on the one hand indicate it visually, and on the other hand disable the button.
 - d. **For an existing grid, make a non-destructive resizing, i.e. keep the actual application state (drawing), and only add or remove the necessary items.
- II. Drawing
 - a. By clicking a cell in the grid, colour the cell to red!
 - b. Store the state of the grid (matrix) in the application state space! On each click store what colour is on that place! Decide whether you display the grid with the imperative or declarative approach!
 - c. *Right-click to delete the colour of that cell (even in the app state)!
 - d. **Make it possible to draw lines! By holding down the mouse button, move the mouse pointer over the grid, and colour those cells that were involved (hovered) during this movement!
- III. Colours
 - a. Store the current colour in the application state space! Use this colour to draw!
 - b. Indicate the current colour in a separate part of the page!
 - c. Create a separate part on the page where there is a colour picker field: `<input type="color">`! If its value changes, then store it in the app state!
 - d. *Under the color picker, show the previously chosen colours as small squares! Click on one of these to set the current colour!
- IV. List of pixel-arts
 - a. Store a lot of pixel-art in the memory (in the state space)! For this choose a suitable data structure! For example:

```

const pixelArts = [
  {
    id: 1,
    pixels: [
      ['#123456', '#234567', '#345678']
    ]
  },
  {
    id: 2,
    pixels: [
      ['#123456'],
      ['#234567'],
      ['#345678']
    ]
  },
]

```

- b. List the previously drawn pixel arts on the page! For displaying these pixel-arts you can generate the same `table` as for the editing, only change the cell size to 1-2px. You can use the `preview` style class specified above!
- c. By clicking on an item, it is loaded for editing: dimensions and the grid. You have to store in the state space that exactly which pixel-art is active!
- d. After editing, clicking a "Save" button replace the old `pixels` matrix in the list to the new one!
- e. Above the list put a "New pixel-art" button! Clicking on this the list gets a new item and this will automatically be selected for editing!
- V. Persistence
 - a. *For each save, save the entire list into the local storage layer of the browser!
 - b. *When loading the page, check if there is any such list in the browser's local storage and load it from there!